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Good Days, Bad Days: Stock Market Fluctuation and Taxi Tipping Decisions

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Abstract. Using taxicab tipping records in New York City (NYC), we develop a novel measure of real-time utility and quantitatively assess the impact of wealth change on the well-being of individuals based on the core tenet of prospect theory. The baseline estimate suggests that a one-standard-deviation increase in the stock market index is associated with a 0.3% increase in the daily average tipping ratio, which translates to an elasticity estimate of 0.3. The impact is short-lived and in line with the wealth effect interpretation. Consistent with loss aversion, we find that the impact is primarily driven by wealth loss rather than gain. We exploit Global Positioning System and timestamp information and design two difference-in-differences tests to establish causal inference. Exploitation of the characteristics of individual stocks suggests that the effect of wealth change on real-time utility is more pronounced in the stocks of firms with large market capitalization. Finally, our aggregate estimate suggests that annual tip revenue in the NYC taxi industry is associated with stock market fluctuations, ranging from –$17.5 million to $12.9 million.

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1. Introduction

A crucial component in neoclassical economic theory is the expected utility framework, wherein consumers make choices based on utilities determined by final outcomes. Kahneman and Tversky (1979) highlight the role of psychological effects and state that the evaluation of an outcome is influenced by its comparison with a reference point and determined by changes from the status quo or differences from the natural reference point.[1] The literature further extends the theoretical framework and develops a generally applicable theory.[2] This reference-dependent preference has two notable manifestations. First, people exhibit a significantly stronger aversion to losses than an appreciation of gains, which is termed *loss aversion* (Tversky and Kahneman 1991). Empirical studies also provide evidence consistent with the loss aversion feature of prospect utility in the housing and stock market (i.e., Barberis et al. 2001, Genesove and Mayer 2001). In addition, the value function is concave in gains and convex in losses, reflecting diminishing sensitivity to changes in an outcome as it moves farther from the reference point.

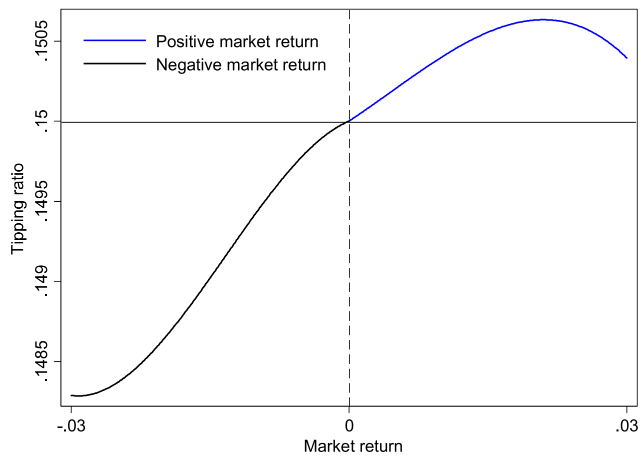
Testing these theories in utility has always been a challenging task for empiricists due to the difficulty in directly observing real-time utility or general well-being of consumers. In experimental settings, certainty equivalence and probability equivalence methods have been the two most frequently used procedures for constructing utility functions and quantifying utility measurement. Researchers have also attempted to infer and measure subjective well-being via surveys.[3] Both approaches that rely on a laboratory or survey setting either hinge on the critical assumption of a consumer's preference consistency, which typically does not correctly reflect a consumer's actual behavior, or are likely to be subject to self-reporting errors common in survey data. Therefore, empiricists take an alternative approach and investigate the impact of wealth change on consumption by exploring various settings as identification.[4]

In this study, we propose a novel and direct measure of real-time utility by exploring etiquette in the service industry, specifically, tipping.[5] Two factors suggest that an examination of tipping records offers a valid measurement of real-time utility. The first factor is the nature of free will, which suggests that people in most situations leave tips voluntarily after a service is rendered. The motivation for consumers to offer tips is primarily psychologically driven and reflects their happiness at the time. Moreover, the actual tipping is highly discretionary because consumers are free to choose how much to tip. We exploit the release of the data set by the New York City (NYC) Taxi and Limousine Commission (TLC) that contains complete trip-level information for all taxicabs in NYC to empirically test this assumption.[6] The data set consists of about 226,000 valid trips per day offered by the NYC TLC and covers all five boroughs in NYC. The richness of our data gives us the opportunity to study tipping variation, a proxy for an individual's real-time happiness, in response to the rise and fall of stock prices. This extensive geographic coverage of NYC and high-frequency nature of the data are crucial to the design of a difference-in-differences (DiD) identification test that can establish causal interpretation.

To better link our proxy for real-time utility to stock market variation, for which daily data are available, we aggregate all taxi trips at the same level and construct a symmetric measure as the daily ratio of tip amount in taxi fares in NYC. This approach enables us to capture variations in both excitement and discontent of a consumer. In their seminal paper on behavioral economics, Kahneman and Tversky (1979) characterize the value function, which describes the decision-making process of individuals as an S-shaped and asymmetric function passing through the reference point. Thus, we first explore how value function is represented in a real-world setting using our taxicab tipping data. We also demonstrate how differences in utility are achieved as a result of certain amounts of gain or loss. Figure 1 provides an illustration. The x-axis indicates the value-weighted market return on a daily basis, and the y-axis plots the daily average tipping ratio (tips/total fares) in NYC. To account for the nonlinear relation, we apply cubic polynomial fitting to plot the curve. Our findings are consistent with prospect theory. First, we observe an S-shaped curve, which indicates that the tipping ratio increases (decreases) as the stock market outperforms (underperforms). In addition, a steeper slope is observed in the negative range than in the positive part with the same magnitude, suggesting that the value function curve is asymmetric and steeper for losses than for gains. This observation supports the loss aversion notion because losses have a greater emotional impact than the equivalent amount of gains.

To generalize these observations into formal tests, we conduct time-series regressions to analyze whether stock market fluctuation has an impact on the well-being of investors, as inferred from tip amounts. We also examine the speed and extent of this impact. Our baseline estimate suggests that a one-standard-deviation increase in the overall market indexes (roughly 1.12%) increases the average tipping ratio in taxicabs by 0.28% on the same day. This estimate translates to an elasticity

Figure 1. (Color online) Stock Market Fluctuations and Tipping



Note. This figure plots asymmetric effects of market returns on tipping behavior. We run the regression as follows from 2009 to 2016:

$$Tipping_t = \alpha + \beta_1 \times Return_t + \beta_2 \times Return_t^2 + \beta_3 \times Return_t^3 + \varepsilon_t,$$

where the dependent variable *Tipping* is the daily ratio of tip amounts on taxi fares in NYC on day t. We run the regression for the negative and positive return days, respectively.

of 0.25, which is comparable with prior studies (i.e., Layard et al. 2008, Boyce and Wood 2011). This response is short-lived and manifests over only the next two days. We also investigate whether the responses exhibit heterogeneity in positive and negative changes of wealth and find that the responses are primarily driven by loss instead of gain. On days with extremely negative returns, individuals cut the tipping ratio by four times more, with a 1% level of significance. However, we do not observe a similar jump in tipping on days with extremely positive returns.

The results remain robust to alternative specifications and explanations. Specifically, the effect is robust to controlling for a wide range of weather conditions (such as snow, clouds, and rain) that can also affect mood (Hirshleifer and Shumway 2003) and account for trips recording zero/default tips. It is still positive and significant when we use Standard & Poor's 500 Index (S&P 500) stock returns as an alternative proxy for stock market performance. Crucially, stock return is measured at the end of the day, whereas taxi rides can occur throughout the day, making the two endogenously determined. Thus, we employ a high-frequency identification strategy based on the exact pickup/drop-off time to show that it is not driven by this look-ahead bias.

Next, we move to differentiate two interpretations of our main finding: (1) *wealth effect*, that is, a reflection of variation in wealth induced by the market fluctuation itself, and (2) *sentiment effect*, namely results driven by market perceptions detached from actual market returns. We adopt two approaches here.

First, we apply the same method as in Engelberg and Parsons (2016) and exclude days with significant event coverage in the *New York Times* and *Wall Street Journal*. The positive effect of stock market returns on tipping is still robust. Second, we construct intraday measures of taxi tipping and market returns and design a high-frequency test to disentangle the two effects. We find that tips react to stock market fluctuations only during trading hours. Taken together, the findings support the wealth-effect hypothesis.

To achieve identification, we implement two sets of tests. First, we use the geographic classifications via the Global Positioning System (GPS) in the data to distinguish treated passengers who are more likely to be affected—stock market participants (investors)—from the rest and draw causal inferences via a DiD test. Second, we further categorize taxi rides into two types, personal and business trips, based on the notion that the latter are less likely to be "the treated" because the costs of business taxi rides are typically borne by the company. Both tests support a causal interpretation for our main results.

Finally, we conduct cross-sectional tests and investigate how stock characteristics could drive the impact of change in wealth on real-time utility. First, we exploit geographic differences across firm headquarters to evaluate portfolio versus nonportfolio effects and find insignificant difference in the impact among stocks of local (NYC-based) and nonlocal firms. In addition, the influence of stock price changes intensifies among stocks with large capitalization. This finding is consistent with the hypothesis that large-cap stocks are likely to receive media coverage and are significant components for an investor's portfolio. To assess the economic significance of the effect described in our paper, we also infer a ballpark number for the total economic gains/losses associated with market fluctuations in the NYC taxi industry. Our estimates imply that annual stock market–induced tip revenue ranges from −$17.5 million to $12.9 million during our sample period.

We contribute to the literature in the following ways. First, we advance the concurrent literature that focuses on outcomes that occur only during downward markets (such as hospital admissions and domestic violence) and is thus unsuccessful in investigating the effect of positive market returns (e.g., Engelberg and Parsons 2016, Lin and Pursiainen 2019). The tipping measure we adopt is symmetric and can capture variations in a consumer's excitement and discontent. Second, we take crucial steps in addressing the identification challenge. With detailed timing and exact GPS information in the data, we can design both high-frequency and DiD tests that help us to achieve identification and obtain a precise estimate of the impact on real-time utility.

We first add to the large literature on tipping.[7] The custom of tipping workers in the service industry dates back centuries ago and is a common practice in many countries. Moreover, its economic magnitude is substantial. For example, annual tipping is estimated to be $46.6 billion in the United States alone (Azar 2011). From the perspective of neoclassical economic theory, tipping appears to be irrational: a utility-maximizing agent should not give up money because the service has already been provided and, more often than not, the agent will not return to the same establishment. This is especially true in industries such as taxicabs, where repeated interactions between customers and taxi drivers are rare. Psychologists and economists alike thus seek to explore the behavioral and psychological reasons for tipping behavior. Researchers have attributed the phenomenon to the role of social norms and emotions (e.g., Strohmetz et al. 2002, Lynn 2015). For instance, Ge (2018) documents that passengers' tipping behavior is a reflection of emotions resulting from sporting event outcomes. Our contribution demonstrates that the consideration of why people tip can be extended to changes in wealth.

Our study also relates to studies that examine micro-level data sets on taxicabs, in particular those investigating the labor-supply decision of cab drivers (i.e., Crawford and Meng 2011, Agarwal et al. 2016). The empirical evidence on the response of labor supply to an increase in transitory wages is at best mixed: some demonstrate negative wage elasticity (i.e., Camerer et al. 1997, Agarwal et al. 2015), whereas others find no significant effects (Farber 2005, 2008). Our study also features real-time and GPS-enabled data, but with a different focus: the tipping behavior of taxi customers.

Last, we are related to the vast literature that tests theories of utility in the stock market setting (e.g., Odean 1998, Barberis et al. 2001, Barberis and Thaler 2003, Ritter 2003, Grinblatt and Han 2005, Baker and Wurgler 2006, Barberis et al. 2006, Kaustia 2010). Specifically, we are among the first few studies to link stock market fluctuations to investor mood using field data, which significantly extends the literature that focuses on experimental settings (Bleichrodt et al. 2010, Engelberg and Parsons 2016, Lin and Pursiainen 2019). Our study differs from these studies by using a symmetric measure that can capture variations in a consumer's excitement and discontent and adopt a better identification strategy. Our approach also complements the literature that explores alternatives to the premise of full rationality and investigates the role of sentiment in asset pricing formation. De Long et al. (1990) present a model that formalizes the role of investors who are not fully rational (i.e., noise traders) in the financial markets. Numerous psychological studies suggest that mood can affect human judgment and behavior (Schwarz and Bless 1991, Frijda and Mesquita 1998). The behavioral finance literature offers evidence of the effects of mood on asset prices (Avery and Chevalier 1999; Kamstra et al. 2000, 2003; Hirshleifer and Shumway 2003; Coval and Shumway 2005). In particular, investor sentiment has explicit normative implications for security markets and can lead to market mispricing (Garcia 2013, Da et al. 2014, Barberis et al. 2016, Li et al. 2019). We investigate the question in a reverse manner and query how stock market fluctuations affect investor mood.

The remainder of this paper is organized as follows. Sections 2 and 3 present the data and summary statistics, respectively. Section 4 reports the main results. Sections 5 and 6 provide additional robustness checks. Section 7 draws our conclusions.

2. Background and Measurements

2.1. NYC Taxi Market

The NYC taxi market is the largest in the United States, with a total of 236 million passenger trips annually and comprising about 34% of all U.S. services (Buchholz 2019). Taxicabs in NYC used to be a distinctive yellow. These yellow NYC taxis were considered icons of the city and were widely used in the arts and media throughout the 20th century. Currently, NYC taxicabs are of two varieties, yellow and green. The latter were recently created as a result of regulatory reform affecting the TLC. To expand taxi services into areas of New York not commonly served by yellow medallion cabs, in the summer of 2013, the TLC initiated the New Boro Taxi Program, also known as the Street Hail Livery Program, and issued licenses for thousands of green Boro taxis to allow New Yorkers to spot a differently colored cab (green taxi) cruising the city streets. The Boro Taxi Program was successful in improving access to street-hailed transportation throughout the five boroughs, especially for people with disabilities and those who live or spend time in areas of NYC that are historically underserved by the yellow taxi industry.[8]

2.2. New York Taxi Data

Taxicabs in NYC are highly regulated by the NYC TLC. Although taxicabs are typically operated by different private companies, taxicab drivers must secure a legal license or a medallion issued by the TLC allowing them to operate. A total of 13,587 taxicab medallion licenses were issued by 2014. Thanks to the Taxi Passenger Enhancement Project implemented by the TLC to mandate the use of upgraded metering and information technology, automated data collection of taxi trip fare information has become possible. In response to a request under New York State's Freedom of Information Law, in 2014, the TLC began the first dissemination of ride-level taxi data (Whong 2014). Although the TLC claims no responsibility for guaranteeing or confirming the accuracy or completeness of the data, it does perform routine reviews of the records and takes enforcement action when necessary to ensure, to the extent possible, completeness and accuracy (NYC TLC 2018). This study is among the first few academic studies (Haggag et al. 2017, Finer 2018, Buchholz 2019) that employ the rich NYC taxi data to provide insight into a broad economic and financial setting.

Each observation in the TLC data denotes a single cab ride. The data include the exact time and date of the pickup and drop-off, the GPS coordinates of the pickup and drop-off,[9] the distance, the passenger count, the fare, any tips, and the manner of payment (cash/card). Because the tipping information for cash payments is not available, our work captures all taxi rides with card payments. However, such a deficiency of the data is helpful for increasing our ability to identify a sample of passengers who are likely to include stocks in their portfolio of wealth (investors). The coordinates are accurate to between $O(10\text{ ft})$ and $O(100\text{ ft})$, and a greater concentration of high rises can introduce bias (Finer 2018). The shortcomings of applying these taxi data in our setting is that we gather no demographic information about the drivers or passengers because the data contain no identifiers or affiliations for them. The key identification assumptions we use are that stock market fluctuations primarily drive the variations in real-time utility of investors and that the estimate can be viewed as the average effect of both investors and noninvestors.

2.3. Taxi Tips and Investor Mood

Our tests require an empirical proxy for the real-time utility or general well-being of investors at any given point in time. We use fluctuations in tipping as a proxy to infer subjective well-being and investor mood. Tipping is a good real-time utility measurement for two reasons. First, tipping is commonly characterized as voluntary and discretionary because the service was already rendered and the consumer is free to choose how much to tip, if at all. Additionally, the reasons people tip are primarily psychological and reflect their feelings at the time. To subject this intuition to empirical testing using the best data available, we exploit the TLC's released data set containing complete trip-level information for NYC.

This measure has several advantages. First, information regarding taxi fare records is not subject to the usual problems of survey data because the data are obtained from technology providers authorized under the Taxicab and Livery Passenger Enhancement Programs. The extensive coverage and comprehensive information in the data (include pickup and drop-off locations, times, fare amounts, and tips) is crucial to our ability to design tests that can identify the influence of stock market fluctuations on investor moods. Moreover, our measure is implicitly symmetric. If a rising market improves the collective mood, we will capture this effect to the extent that tipping increases, whereas a downward market deteriorates investor mood, thereby decreasing tips.

3. Sample and Methodology

3.1. Sample

The taxi trip sample used in this study is a relatively complete version of the TLC data set from January 2009 to December 2016. We apply a light filtration process and exclude trips that appear to be duplicates or that do not meet reasonable conditions and include trips with total payments and tip amounts that are positive but less than $1,000, at least one but fewer than five passengers, and pickup and drop-off locations in NYC. To maximize the probability that our estimate captures the response to market performance of the event day, in the main test, we only consider trips after 9:30 a.m., when regular market trading starts. The growth in popularity of ride-hailing apps such as Uber has affected NYC's cab industry ridership numbers over the past several years. Statistics based on Certify data indicate that these services' share of expensed rides increased from roughly one-tenth in early 2014 to roughly one-third in early 2015 (White 2015). To mitigate concerns regarding the impact of Uber popularity on tipping behavior, we also choose an earlier endpoint, 2014, and check the robustness of our main estimates.

We extract stock prices and return data from the Center for Research in Security Prices (CRSP) and firm location information from Compustat. Then we merge the two data sets using the new common CRSP–Compustat link file. Compustat provides a five-digit ZIP code for each firm's headquarters, which we use to classify whether the firm is headquartered in NYC or not. To better link investor tipping behavior to daily stock market indexes, we aggregate taxi trips at the daily level and merge the aggregated taxi trip record with the return data for the same day. For example, for market returns on day t (i.e., April 25, 2014), which will always be a trading day, the average daily tipping on day t will be that on April 25, 2014, whereas days $t+1$ and $t+2$ correspond to Friday, April 26, and Saturday, April 27, 2014, respectively. In other words, day $t+n$, on which tipping is averaged to occur, might not always be a trading day.

Table 1 reports the summary statistics of our variables of interest. During our sample period, the mean and median ratios of tips to trip fares in NYC are 14.99% and 15.16%, respectively, which are comparable with the observations of previous studies (Camerer et al. 1997, Farber 2008). On average, the distance driven per trip is approximately three miles, and the taxi fare for each trip is about $16.02. Our aggregated version of the daily tipping variable is calculated based on an average of 225,657 trips. Average daily U.S. stock returns are about 6 basis points (bps), whereas those of S&P 500 stocks are about 5 bps per day. During our sample period, the stocks of firms headquartered in NYC offer a high but volatile return, with an average return of 8 bps and a standard deviation of 128 bps per day.

3.2. Methodology

First, we test for a contemporaneous relation between stock market performance and taxi tipping behavior by estimating the regression for all trading days t from January 2009 to December 2016.

$$\ln(Tip\ ratio)_t = \alpha + \beta \times Return_t + \gamma \times \boldsymbol{Control}_t + \varepsilon_t, \tag{1}$$

where the dependent variable $\ln(Tip\ ratio)_t$ is the natural logarithm of the daily ratio of tip amounts to taxi fares in NYC on day t. We aggregate the records for all taxi trips after trading hours to obtain the average. We also consider an alternative aggregation method based on trip characteristics, such as the pickup time and area. The variable $Return_t$ denotes stock market performance (value-weighted indexes) on day t. To facilitate the interpretation of the coefficient, our benchmark regressions scale the daily value-weighted market stock return by the time-series standard deviation. We present the results for alternative measures, including *S&P 500 return* and *NYC stock return* in the robustness check. We are primarily interested in the coefficient of interest β, which is an estimate of the contemporaneous relation between stock market performance and tipping behavior.

In the vector $\mathbf{Control}_t$, we employ a list of variables to control for the potential temporal patterns of tipping behaviors. First, we include year–month fixed effects to account for long-term variation in customer mood and other changes in wealth that can affect tipping behavior. We also control for seasonality within the same year and weekday by adding fixed effects for the days of the week. Finally, we include indicator variables for the three days surrounding each of the following holidays: New Year's Day, Fourth of July (Independence Day), Labor Day, Thanksgiving, and Christmas. Thus, we control for holiday effects because several studies find that people are more generous during holiday periods (Wallendorf and Arnould 1991). Prior studies indicate that weather conditions can also affect mood, raising it with exposure to bright light and lowering it with decreased sun exposure.

Table 1. Summary Statistics

Variable	Mean	Standard deviation	25th percentile	Median	75th percentile
Daily average taxi tip amount	2.4233	0.4572	2.1963	2.4126	2.6406
Daily number of pickups	225,657.8	64,088.5	182,158	237,990	274,704
ln(*Tip ratio*)	2.7071	0.0620	2.6958	2.7101	2.7213
ln(*Total taxi fares*)	2.8362	0.0959	2.7437	2.8593	2.9234
Daily U.S. return	0.0006	0.0112	−0.0041	0.0008	0.0058
Daily S&P 500 return	0.0005	0.0110	−0.0041	0.0006	0.0057
Daily NYC return	0.0008	0.0128	−0.0049	0.0010	0.0068

Notes. The *Daily average taxi driver tip amount* is the daily average of tips taxi drivers receive. The *Daily number of pickups* is the number of pickups daily. The variable ln(*Tip ratio*) is defined as the natural logarithm of daily average tip ratio, in which the daily average tip ratio is defined as (the sum of tips divided by sum of total fares) × 100 based on the trip data after trading hours starts (9:30 a.m.) in a given day. The variable ln(*Total taxi fares*) is defined as the natural logarithm of the average daily taxi fare. The *Daily U.S. return* is the value-weighted daily return of U.S. stocks. The *Daily S&P 500 return* is the value-weighted daily return of S&P 500 stocks. The *Daily NYC return* is the value-weighted daily return of stocks whose firm is headquartered in NYC.

This effect could simultaneously lead to variations in an individual's tipping behavior and have asset pricing implications (Hirshleifer and Shumway 2003). We also control for weather condition fixed effects (with dummy variables indicating whether the conditions are snowy, cloudy, or rainy) for a robustness check.

To explore the dynamics of tipping behavior responses, we replace $\ln(\mathit{Tip\ ratio})_t$ with a lead–lag version $\ln(\mathit{Tip\ ratio})_{t+\varsigma}$. This empirical specification with positive values for ς allows us to investigate whether investors exhibit delayed awareness because prior studies on investor inattention suggest that they tend to have limited amounts of time and cognitive resources with which to process information (Della Vigna and Pollet 2009).[10] Our empirical strategy seeks to test utility theory by establishing the impact of wins/losses on the real-time utility of investors. However, the underlying mechanism might work in the other direction because investor sentiment can largely affect future stock prices. For example, Baker and Wurgler (2006) suggest that beginning-of-period proxies for sentiment strongly predict subsequent returns for stocks and affect the cross section of stock returns. The distributed lag model helps us test for a lagged relation between tipping behavior and our proxy for real-time utility stock market performance and exclude such possibilities. The appendix presents the definitions of the variables used in the test.

4. The Impact of Stock Market Fluctuations and Investor Mood

4.1. Baseline Estimation

We begin by estimating the contemporaneous impact of stock market performance on the tipping behavior of investors, as in Equation (1). Table 2 reports the estimated coefficients from these regressions, where we progressively add control variables across columns. Column (1) indicates a positive and significant relation between stock market performance and the concurrent tipping ratio, which controls for year, month, and day-of-the-week fixed effects. The point estimate is 30.20 bps and is statistically significant at the 5% level. The observed impact on tipping behavior could reflect the holiday effect, implying that people are more generous because it is a holiday period (i.e., Greenberg 2014). The level of total taxi fares can also affect tipping variations. Thus, we include holiday fixed effects and a natural logarithm of the average amount of taxi fares in columns (2) and (3). Adding additional controls neither affects this relation nor changes its level of significance. The coefficient estimate in column (3) is 0.276 and still statistically significant ($p < 0.05$). The stability of estimates suggests that omitted variable issues would not be cause for concern if one assumes that any unobservable characteristics are similar to the observable characteristics. The economic magnitude of the estimate is also significant and consistent with what we observe in Figure 1. For example, the coefficient estimate in column (3) indicates that an increase of one standard deviation, which is roughly a

Table 2. Market Fluctuations and Taxi Tipping: Baseline Results

Dependent variable: ln(*Tip ratio*)	(1)	(2)	(3)
Market return	0.302**	0.306**	0.276**
	(0.04)	(0.03)	(0.05)
ln(*Total taxi fares*)			−0.772***
			(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	No	Yes	Yes
N	2,015	2,015	2,015
Adjusted R^2	0.047	0.047	0.127

Notes. The main independent variable is the daily *Market return*, which is scaled by the sample standard deviation. The sample period is 2009–2016. The p-values adjusted by robust standard errors (White 1980) are in parentheses.

$^{**}p < 0.05$; $^{***}p < 0.01$.

1.1% increase in the stock market return, increases the daily tipping ratio by 0.28%.

4.2. Dynamics

To sharpen our strategy, we analyze the dynamics of tipping responses to stock market performance using the lead–lag model. Our prior findings can occur when investors exhibit limited attention and are aware of movements in stock prices over the course of a few days after the event date. It is also possible that investors' attention closely tracks stock market performance, but their generosity is delayed. Thus, we expect a positive and significant relation between stock market performance and tipping behavior the following day, whereas no relation is observed between stock market performance and the previous day's tipping. Tipping is an individual behavior, and generosity is simply a proxy for market sentiment reflecting an optimistic market atmosphere. Thus, our findings in Table 2 could suggest evidence of investors' prediction of future stock market performance.

Our tests allow us to distinguish between these possibilities and establish a causal effect. We characterize the lead–lag relation, allowing both past (days $t-2$ and $t-1$) and future (days $t+1$ and $t+2$) stock market variables to influence taxi tipping on day t. We find that the effect of stock market fluctuations on tipping shows up on the same day as well as the next day.

Panel A of Table 3 presents the estimate when we stack the returns on different days together to integrate all timing-related tests into one regression, whereas panel B examines the lead–lag relation on a one-by-one basis. The findings presented in Table 3 consistently reject the conjecture that tipping leads to stock market performance. This finding implies that the effect of stock market fluctuations on tipping behavior is primarily front loaded because the response is concentrated over days t and $t+1$. Alternatively, the impact on tipping before event day t or after $t+1$ is indistinguishable from zero. Economically speaking, a one-standard-deviation decline in stock market returns increases the average tipping ratio by about 0.276% and 0.201% on days t and $t+1$, respectively.

4.3. Robustness

We perform a battery of alternative tests in this section to evaluate the robustness of the statistical association between stock market performance and the real-time utility of investors.

4.3.1. Alternative Specifications.

We begin by recognizing that value-weighted market returns are not publicly disclosed in real time and that investors may be aware of alternative well-constructed stock market indexes. As a solution to the attention issue, we use S&P 500 stock returns as an alternative proxy for stock market performance. As shown in column (1) of panel A of Table 4, this alternative measure does not yield results that are qualitatively or quantitatively different from our baseline estimates. Column (2) of panel B reports the results that further control for weather condition fixed effects. We find consistent results.

4.3.2. Zero Tips.

In our sample, we find that, on average, 3% of the trips paid with a credit card are recorded as zero tips. As a robustness check, we repeat the baseline test using the ratio of total tips to total taxi fares on a given day, excluding all zero-tip trips, and tabulate the results in column (3) of panel A of Table 4. We find quite similar results: the coefficient on market return is positive and significant at the 5% level.[11]

4.3.3. Look-Ahead Bias.

Given that the stock return is measured at the end of the day, whereas taxi rides can occur throughout the day, the look-ahead bias might rise; that is, stock returns, information, and the number of taxi rides are endogenous. We address this concern by taking advantage of the detailed timing data and employing a high-frequency identification strategy based on the exact pickup/drop-off times for each taxi ride.

Specifically, we obtain hourly S&P 500 returns from Tick Data and aggregate them in the morning sessions (9:30–11:30 a.m.) and afternoon sessions (2–4 p.m.) of trading for each day. Then we construct the tipping variables ln(*Tip ratio*) and ln(*Total taxi fares*) for the time slots for lunch (11:30 a.m.–2 p.m.) and dinner (6–10 p.m.), respectively. We then examine the impact of market returns over different past horizons (morning and afternoon) on future tipping behavior. If our result is free of look-ahead bias, we would expect taxi tipping during lunchtime to be positively affected by morning-session market returns, whereas dinnertime tipping is only affected by afternoon-session returns and not by morning sessions.

Panel B of Table 4 presents the results. In column (1), where we relate the first half-day's stock returns to the tipping behaviors of taxi rides during lunchtime, we find that the coefficient estimate over morning market returns is positive and economically large, although statistically insignificant. However, when we further check its effect on tipping behavior over dinnertime in column (2), the estimate is small and negative. In column (3), we confirm that afternoon-session market returns can significantly and positively affect dinnertime taxi tipping.

4.3.4. Other Taxi Ride Characteristics.

Our interpretation of the relations between the stock market and taxi tipping is based on an implicit assumption that the same/similar passengers are taking taxis on days with different market returns. However, the assumption

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Table 3. Market Fluctuations and Taxi Tipping: Dynamics

Panel A: Lead–lag relation in stacked test

Dependent variable: ln(*Tip ratio*)	(1)	(2)
Return of day $t + 2$	0.087	0.076
	(0.55)	(0.58)
Return of day $t + 1$	0.280*	0.282**
	(0.06)	(0.04)
Return of day t	0.324**	0.295**
	(0.03)	(0.03)
Return of day $t - 1$	0.022	0.059
	(0.88)	(0.67)
Return of day $t - 2$	−0.041	−0.015
	(0.78)	(0.91)
Ln(Total taxi fares)		−0.775***
		(0.00)
Day-of-the-week fixed effects	Yes	Yes
Month × year fixed effects	Yes	Yes
Holiday fixed effects	Yes	Yes
N	2,013	2,013
Adjusted R^2	0.047	0.128

Panel B: Lead–lag relation in separate test

Dependent variable: ln(*Tip ratio*)	(1) Day $t - 2$	(2) Day $t - 1$	(3) Day t	(4) Day $t + 1$	(5) Day $t + 2$
Market return	−0.040	0.027	0.276**	0.201**	0.078
	(0.77)	(0.85)	(0.05)	(0.05)	(0.58)
ln(*Total taxi fares*)	−0.666***	−0.673***	−0.772***	−0.682***	−0.634***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes	Yes	Yes
N	2,013	2,014	2,015	2,015	2,015
Adjusted R^2	0.112	0.112	0.127	0.116	0.108

Notes. Panel A presents the estimate when we stack the returns on different days together, whereas in panel B we examine the lead–lag relation separately. The main independent variable is the daily *Market return*, which is scaled by the sample standard deviation. The sample period is 2009–2016. The *p*-values adjusted by robust standard errors (White 1980) are in parentheses.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

can be violated because stock market fluctuations could influence an individual's decision of whether to take a taxi ex ante. Thus, we further describe the characteristics of taxi riding behavior to validate this assumption and exclude alternative potential explanations.

We first conduct a univariate comparison of trip-related variables on days with good versus bad market performance, including the number of trips by taxi, the likelihood of a cash versus a card payment, ride distance, the number of passengers per pickup, and the proportion of zero-tip payments. The results are tabulated in Table A2 of the online appendix. We compare the trip characteristics on good days versus bad days in univariate analysis (panel A) and multivariate regression (panel B). All these variables are similar between good and bad market return days because the *t*-values show no significant differences between the two groups. The conclusion holds when we contrast the median values and implement *z*-tests.

To better understand how the geographic composition of passengers varies with daily fluctuations of stock market performance, we also present visual evidence by plotting the heat map of the pickup/drop-off locations on good and bad days. Figure 2 presents the distributions of the pickup and drop-off locations of the top and bottom 10% market return days. The figures show no obvious evidence of differences in the pickup/drop-off locations of taxi rides between good and bad days, supporting the implicit assumption.

4.4. Asymmetric Effects of Positive and Negative Market Performance

In this section, we test whether the magnitude of the effect on taxi tipping varies for positive and

Table 4. Market Fluctuation and Taxi Tipping: Robustness

Panel A: Alternative variables and model specifications

Dependent variable: ln(*Tip ratio*)	(1) S&P 500 return	(2) Controlling for weather effects	(3) Dropping zero tip trips
Market return	0.272**	0.269*	0.287**
	(0.05)	(0.05)	(0.04)
ln(*Total taxi fares*)	−0.772***	−0.774***	−0.794***
	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
Weather condition fixed effects	No	Yes	No
N	2,015	2,015	2,015
Adjusted R^2	0.127	0.127	0.151

Panel B: Look-ahead bias

Dependent variable: ln(*Tip ratio*)	(1) 11:30 a.m.–2 p.m. Lunchtime	(2) 6–10 p.m. Dinnertime	(3) 6–10 p.m. Dinnertime
Return from 9:30 to 11:30 a.m.	0.214	−0.070	
	(0.27)	(0.63)	
Return from 2 to 4 p.m.			0.360**
			(0.02)
Corresponding ln(*Total taxi fares*)	−0.560***	−1.072***	−1.066***
	(0.00)	(0.00)	(0.00)
Intercept	2.906***	3.012***	3.010***
	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
N	2,015	2,015	2,015
Adjusted R^2	0.765	0.870	0.870

Notes. In panel A, column (1), the independent variable *S&P 500 return* is the value-weighted daily return of S&P 500 stocks. In column (2), we further control for weather condition fixed effects (with the dummy variables *Snowy*, *Cloudy*, and *Rainy*). In column (3), we drop trip fares for which a zero tip is recorded and generate the natural logarithm of daily average tips for taxi fares. The p-values adjusted by robust standard errors (White 1980) are in parentheses.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

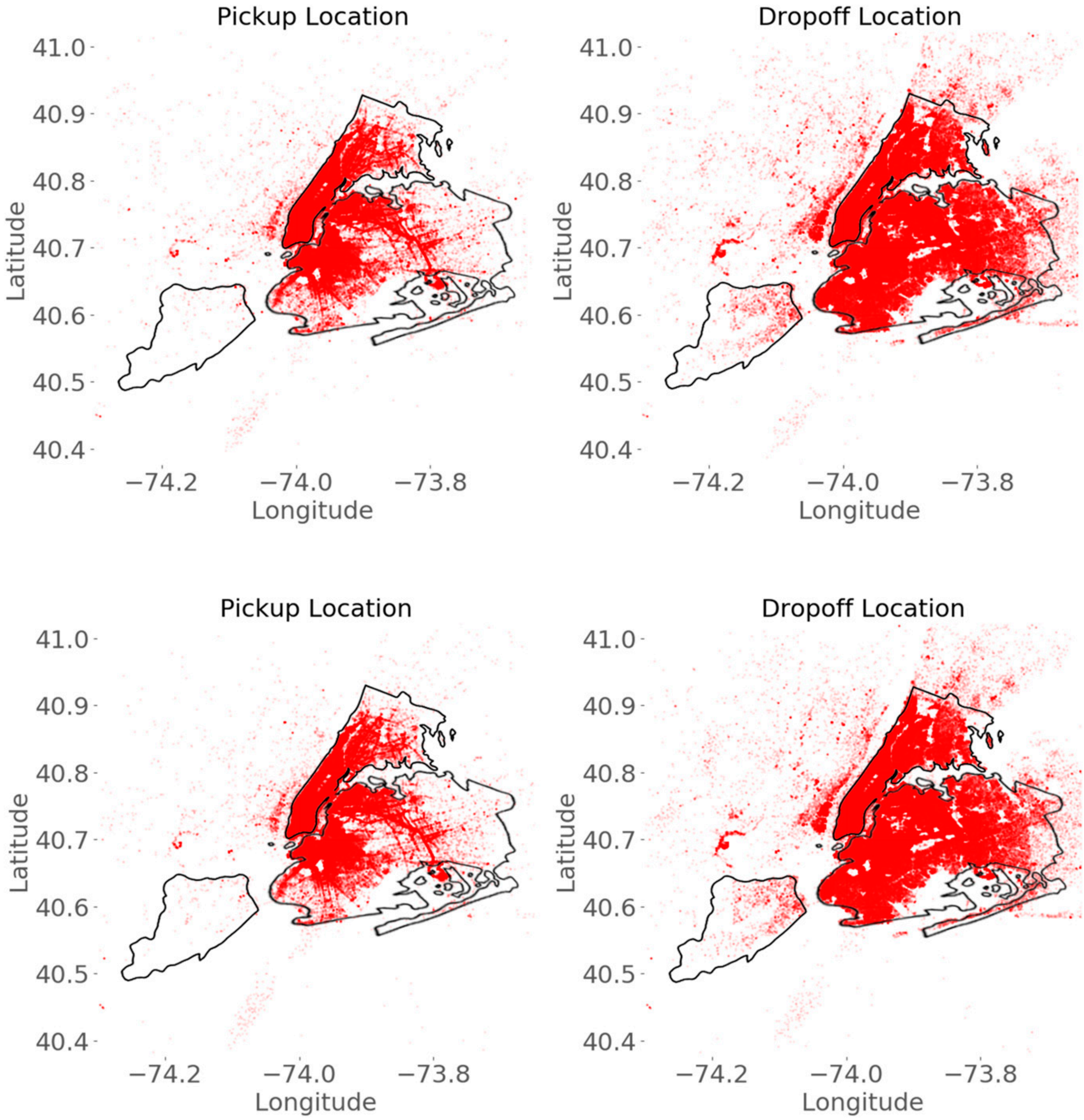
negative market returns. We estimate the following regression:

$$\ln(Tip\,ratio)_t = \alpha + \beta_1 \times Negative\,return_t + \beta_2 \times Positive\,return_t + \gamma \times \boldsymbol{Control}_t + \varepsilon_t, \tag{2}$$

where the dependent variable ln(*Tip ratio*) is the natural logarithm of the daily ratio of tipping amount to taxi fares in NYC.[12] Stock market return information is included in a manner that reflects the wealth gain/loss. The variable *Negative return* is equal to the stock market return if the stock market return is negative and zero otherwise. The variable *Positive return* is equal to the stock market return if the stock market return is positive and zero otherwise. Both stock returns are scaled by their rolling standard deviation. An asymmetry effect predicts that the coefficient β_1 should be significantly larger than β_2. Table 5 presents the results. In column (1), we only include the variable *Negative return* in the regression and find that the coefficient β_1 is 0.686 and significant at the 1% level. However, we find that the coefficient β_2 is insignificant when we include the variable *Positive return* in the regression in column (2). When both *Negative return* and *Positive return* enter the specification in column (3), the effect of *Negative return* becomes dominant because only the coefficient β_1 is positive and significant. These findings offer evidence consistent with prospect theory, indicating that an individual's real-time utility is asymmetrically dependent on stock market performance.

We extend this analysis further by examining how extreme market returns affect the tipping behavior of investors. We adopt a similar specification as in

Figure 2. (Color online) Distribution of Pickup and Drop-Off Locations on Good **vs.** Bad Days



column (3) of Table 2 but replace the continuous return variable with four dummies that represent the bottom quintile, quintiles 2 and 4, and the top quintile of the empirical distribution, respectively. The results again appear to suggest that an individual's real-time utility, as proxied by his or her tipping behavior, is primarily driven by the wealth loss rather than the gain. When the market return is in the bottom quintile of the distribution, taxi tips decrease by 1%. Interestingly, we did not find a significant impact of top-market return quintiles on investor tipping.

We further examine the impact of return saliency because a large return (either extremely positive or extremely negative) could affect whether consumers check their stock portfolios more often. Thus, following extreme returns in the past couple of days, tips may be temporarily more sensitive to stock market fluctuations. Given that the impact on tipping is asymmetric and mostly driven by the bottom quintile of market returns, this result is particularly relevant for us in understanding the dynamics of the effect of extreme returns on taxi tipping.

Table 5. Market Fluctuations and Taxi Tipping: An Asymmetric Effect

Panel A: Negative vs. positive returns

Dependent variable: ln(*Tip ratio*)	(1)	(2)	(3)
Negative return	0.686***		0.812***
	(0.00)		(0.00)
Positive return		0.073	−0.304
		(0.76)	(0.26)
ln(*Total taxi fares*)	−0.773***	−0.774***	−0.774***
	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
N	2,015	2,015	2,015
Adjusted R^2	0.130	0.125	0.130

Panel B: Extreme returns

Dependent variable: ln(*Tip ratio*)	(1)	(2)	(3)
Market return: *Bottom quintile*	−0.009***		−0.010**
	(0.01)		(0.02)
Market return: *Quintile 2*			−0.001
			(0.83)
Market return: *Quintile 4*			−0.000
			(0.97)
Market return: *Top quintile*		0.000	−0.003
		(0.91)	(0.52)
ln(*Total taxi fares*)	−0.773***	−0.774***	−0.774***
	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
N	2,015	2,015	2,015
Adjusted R^2	0.129	0.125	0.127

Panel C: Saliency of extreme returns

Dependent variable: ln(*Tip ratio*)	(1) Day $t-2$	(2) Day $t-1$	(3) Day t	(4) Day $t+1$	(5) Day $t+2$
Market return × *Bottom quintile*	−0.153	−0.257	0.918*	0.863*	−0.008
	(0.76)	(0.61)	(0.06)	(0.08)	(0.99)
Market return	0.003	0.002	−0.000	0.003	0.001
	(0.69)	(0.79)	(0.99)	(0.66)	(0.90)
Bottom quintile	0.147	0.292	−0.173	−0.019	0.014
	(0.52)	(0.20)	(0.44)	(0.93)	(0.95)
ln(*Total taxi fares*)	−0.684***	−0.730***	−0.774***	−0.703***	−0.712***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes	Yes	Yes
N	2,013	2,014	2,015	2,015	2,015
Adjusted R^2	0.112	0.118	0.129	0.112	0.113

Notes. In panel A, the variable *Negative return* is equal to the stock return if the stock return is negative and zero otherwise. The variable *Positive return* is equal to the stock return if the stock return is positive and zero otherwise. In panel B, the variable *Bottom quintile* is equal to one if the daily stock return is in the bottom quintile and zero otherwise. The variable *Top quintile* is equal to one if the daily stock return is in the top quintile and zero otherwise. In panel C, the variable of interest is the interaction between *Bottom quintile* and *Market return*. The market return is scaled by the sample standard deviation. The sample period is 2009–2016. The p-values adjusted by robust standard errors (White 1980) are in parentheses.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 6. Disentangling the Wealth from the Sentiment Effect: Evidence from Periods of No News

Dependent variable: ln(*Tip ratio*)	(1)	(2)	(3)
Bottom quintile: All	−0.009***		
	(0.01)		
Bottom quintile: No wars/terrorist attacks		−0.010***	
		(0.01)	
Bottom quintile: No news			−0.015***
			(0.00)
ln(*Total taxi fares*)	−0.773***	−0.773***	−0.777***
	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
N	2,015	2,015	2,015
Adjusted R^2	0.129	0.129	0.135

Notes. The main independent variable in the table is the bottom quintile of market returns. Column (1) reproduces column (1) in Table 5, panel B. Column (2) is identical to column (1), except the bottom quintile dummy equals one if the returns are in the bottom quintile and there is no news of a war or terrorist attack and zero otherwise. Column (3) is identical to column (1), except the bottom quintile dummy equals to one if returns are in the bottom quintile and there is no news event and zero otherwise. The sample period is 2009–2016. The p-values adjusted by robust standard errors (White 1980) are in parentheses.

***$p < 0.01$.

We first create an indicator to highlight days with extreme negative returns, that is, *Bottom Quintile*, a dummy that is equal to one if the daily stock return is in the bottom quintile and zero otherwise. Then we run a lead–lag regression as in our dynamic test and consider the predictability of extreme negative market returns on day t for tipping on days $t-2$ to $t+2$ by interacting the market return with the bottom quintile. The results are summarized in panel C of Table 5. We find that the coefficients over the interaction term are significantly positive on days t and $t+1$, implying that passengers' tipping behavior indeed becomes more sensitive to fluctuations in the stock market following extremely negative market returns, but the pattern is short-lived.

4.5. Disentangling the Wealth vs. Sentiment Effect

In this section, we conducts tests to examine two interpretations of our main findings: (1) wealth effect, that is, wealth effects generated by the market itself, and (2) sentiment effect, namely the results are driven by market perceptions, detached from actual market returns. To account for factors related to the perception of the stock market or overall sentiment embedded in the news, we apply the same approach as Engelberg and Parsons (2016). Because the bottom quintile returns are entirely responsible for the relation between market returns and taxi tipping behavior, we collect the news items in the *New York Times* and *Wall Street Journal* on days with returns in the bottom quintile. We classify all events into the following categories: (1) wars or terrorist attacks, (2) macro announcements, (3) firm announcements, (4) prices in other markets, and (5) other events.

In our bottom quintile sample, wars and terrorist attacks are reported in 12 cases, whereas no news event is provided in 57.8% of cases. We rerun the specification shown in panel B of Table 5 but exclude certain types of news. Table 6 presents the results. Column (1) of Table 6 reproduces column (1) of panel B of Table 5. In column (2), we exclude the 12 days that involved a war or terrorist attack in the news. These events are arguably the most capable of independently affecting investor well-being and are often accompanied by extreme market declines. Accordingly, if their exclusion meaningfully alters our results, the explanation of our main results would become suspect. As can be seen, however, the coefficient is nearly identical to that in column (1), suggesting that these days play virtually no role in the main effect of market returns on taxi tipping behavior. In this column, we try to provide evidence that our main results are driven at least in part by the market return per se rather than the news it reflects. We define the key variable of interest, the bottom quintile dummy, to be one only if the returns are in the bottom quintile and there is no news event. The coefficient is −0.015 and significant at the 1% level. The results point to a wealth-based interpretation of our main finding.

Taking advantage of the granularity of the taxi data, we further construct intraday measures and use the timing of the taxi rides and market opening hours to design a better test. Specifically, we introduce intraday return and S&P 500 Futures Index return tests to

Table 7. Disentangling the Wealth from the Sentiment Effect: Intraday Evidence

Panel A: Intraday evidence

Dependent variable: ln(*Tip ratio*)	(1) 7:30–9:30 a.m.	(2) 9:30–11:30 a.m.	(3) 7:30–9:30 a.m.
Overnight market return	0.048	0.457**	
	(0.80)	(0.01)	
Return ($Close_{t-1} - Open_{t-1}$)			0.518***
			(0.00)
Corresponding ln(*Total taxi fares*)	−0.587***	−0.736***	−0.599***
	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
N	2,015	2,015	2,015
Adjusted R^2	0.767	0.769	0.768

Panel B: Evidence using S&P 500 Futures Index returns

Dependent variable: ln (*Tip ratio*)	(1) 6–10 p.m.	(2) 6–10 p.m.
Return of S&P 500 Futures Index, 2–4 p.m.	0.341**	
	(0.02)	
Return of S&P 500 Futures Index, 4–6 p.m.		0.047
		(0.75)
ln(*Total taxi fares*)	−1.067***	−1.071***
	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes
Month × year fixed effects	Yes	Yes
Holiday fixed effects	Yes	Yes
N	2,015	2,015
Adjusted R^2	0.870	0.870

Notes. Sample period is 2009 to 2016. The p-values adjusted by robust standard errors (White 1980) are in parentheses.
** $p < 0.05$; *** $p < 0.01$.

further disentangle the wealth effects. The logic is that if the tips are indeed reactions to market drops, then news that could potentially cause a drop in the market should not affect tips during nontrading hours.

Close-to-open returns incorporate news that was released during the night, and this would show up in observable prices only after the stock market opens at 9:30 a.m. Therefore, close-to-open returns should not be associated with taxi tipping from 7:30 to 9:30 a.m. but should be associated with taxi tipping from 9:30 to 11:30 a.m. Furthermore, the previous day's open-to-close returns should be associated with tipping from 7:30 to 9:30 a.m. Consequently, we could rule out the potential concern that morning commute tipping is just generally insensitive.

To implement such tests, we use the data from Tick Data, which provides real-time daily market returns and S&P 500 futures trading data. We examine the effect of intraday returns on taxi tipping and report the results in Table 7. In columns (1) and (2) of panel A, we construct the overnight market return by calculating the return of the close index of day $t - 1$ to the open index of day t. We construct a variable for taxi tipping from 7:30 to 9:30 a.m. on day t. Column (1) shows that the coefficient on the overnight return variable is insignificant. We further construct the variable taxi tipping from 9:30 to 11:30 a.m. on day t. Column (2) show that the coefficient on the overnight return variable is positive and significant, implying that taxi passengers observe and respond to close-to-open market returns incorporating news that was released during the night only when the market opens. Further study investigates whether taxi passengers before the morning trading hour respond to the market return of day $t - 1$. Column (3) shows that the previous day's open-to-close returns are associated with tipping from 7:30 to 9:30 a.m. Thus, the result rules out potential concern that morning commute tipping is just generally insensitive.

Alternatively, we design a cleaner test using movements in S&P 500 futures, which continue to trade even after the stock market closes. We believe that the S&P 500 Futures Index is less likely to draw attention from ordinary individuals. Column (1) in panel B of

Table 7 shows that the S&P 500 futures returns from 2 to 4 p.m. negatively affect the taxi tipping of evening passengers (from 6 to 10 p.m.) at the 5% level, whereas column (2) shows that postmarket close S&P 500 futures returns from 4 to 6 p.m. do not matter for tipping in the evening. The results offer support for our interpretation that stock market movement affects tipping, not just news.

5. Two Identification Tests

One concern about our previous results is that tipping behavior and stock market performance could be spuriously correlated because of unobserved regional characteristics or reverse causality. In this section, we formally address this issue using two identification tests.

5.1. Wall Street Area

The results in Section 4 confirm that fluctuations in stock market returns affect the tipping behavior of investors at the aggregate level. Given that the data do not allow us to observe the demographics for each customer, we rely on the following two assumptions: (1) stock market fluctuations reflect changes in investor wealth, and (2) the aggregated tipping ratio is representative of investors.[13] The former is reasonable, but the latter could suffer from severe measurement error; that is, treating all taxi customers as investors in the stock market could introduce an upward bias to our estimates.

To alleviate concerns about the validity of our empirical strategy, we add another cross-sectional dimension and design a DiD test to obtain a precise estimate of the impact. The GPS allows us to conduct a finer geographic classification to identify treated passengers who are more likely to be affected—that is, stock market participants (investors)—from the rest and draw causal inference via a DiD test. Empirically, we take advantage of the GPS information in the taxi ride database to identify the location of pickups and drop-offs and compare differences in the tipping behavior of passengers who are financial industry employees versus that of others.

Thus, we refine the classification by first identifying passengers who were picked up or dropped off around Wall Street. These passengers are more likely to work in the financial sector, and their tipping behavior should respond more to stock market fluctuations. We construct a dummy variable *Wall Street* that takes the value of one if the pickup or drop-off location is within 100 meters of Wall Street and zero otherwise. To obtain a valid counterfactual, we consider similar passengers who work in other industries by identifying taxi rides whose pickup or drop-off location is in midtown Manhattan, the area between 34th Street and 59th Street, bounded by the East River to the east and the Hudson River to the west.[14]

We apply a DiD approach to establish the causality. Specifically, we examine changes in the tipping behavior of the treatment group (versus that of the control group) induced by stock market fluctuations using the following specification:

$$\begin{aligned}\ln(\mathit{Tip\ ratio})_{t,i} = \alpha &+ \beta_1 \times \mathit{Wall\ Street}_{t,i} \\ &+ \beta_2 \times \mathit{Wall\ Street}_{t,i} \times \mathit{Return}_t \\ &+ \gamma \times \mathit{Controls}_t + \varepsilon_{t,i}, \end{aligned} \tag{3}$$

where the dependent variable $\ln(\mathit{Tip\ ratio})_{t,i}$ is the natural logarithm of the daily average tipping for group i on day t. The variable $\mathit{Wall\ Street}_{t,i}$ is our treatment indicator, and it takes the value of one if the pickup or drop-off location is within 100 meters of Wall Street and zero if the pickup or drop-off location falls in midtown Manhattan. In the regression, we control for ln(*Total taxi fares*) on the day-trip-type level and also day fixed effects.[15] Standard errors are clustered by day and location. The coefficient of interest is β_2 for the interaction term $\mathit{Wall\ Street}_{t,i} \times \mathit{Return}_t$, which captures the difference in tipping behaviors between the treatment and control groups as induced by stock market fluctuations.

Panel A of Table 8 shows the results where we focus on the impact of S&P 500 and value-weighted market returns. We find that the coefficients on the interactions between *Wall Street* and market returns are significantly positive, and the magnitude is larger than in the benchmark specification (0.276). The finding that stock market participants are more sensitive to market fluctuations is in line with the causal interpretation of our main results.

5.2. Personal vs. Business Trips

Because the costs of business taxi rides are typically borne by the company rather than the individual, we expect a stronger effect for trips that are likely to be personal trips that those likely to be business trips. Thus, we further categorize the taxi rides into two types, personal and business trips, by taking advantage of the GPS location and time information. We obtain the longitude and latitude of firms' headquarters through their business addresses in the Compustat database. Then we identify the taxi ride as a business trip if the pickup or drop-off location falls within 100 meters of the headquarters of listed companies in NYC and the trip time is between 7:30 a.m. and 6 p.m. Thus, we classify taxi rides into a business trip group and generate the variables ln(*Tip ratio*) and ln(*Total taxi fares*) for each trading day.

To obtain a clean sample of counterfactuals, we do not include the rest of the taxi rides (those with pickup or drop-off locations away from the headquarters of

Table 8. Two Identification Tests

Panel A: Wall Street vs, other trips		
Dependent variable: ln(*Tip ratio*)	(1)	(2)
Wall Street × *S&P 500 return*	0.915**	
	(0.05)	
Wall Street × *Market return*		0.841*
		(0.06)
Wall Street	–0.601***	–0.601***
	(0.00)	(0.00)
ln(*total taxi fares*)	1.433***	1.434***
	(0.00)	(0.00)
Day fixed effects	Yes	Yes
N	3,774	3,774
Adjusted R^2	0.728	0.728

Panel B: Business vs. personal trips		
Dependent variable: ln(*Tip ratio*)	(1)	(2)
Personal trip × *Market return*	0.549*	
	(0.08)	
Personal trip × *S&P 500 return*		0.551*
		(0.08)
Personal trip	–0.504***	–0.504***
	(0.00)	(0.00)
ln(*Total taxi fares*)	–0.586***	–0.586***
	(0.00)	(0.00)
Day fixed effects	Yes	Yes
N	3,774	3,774
Adjusted R^2	0.926	0.926

Notes. The main independent variable in this table is the daily stock return, which is the S&P 500 return or market return, respectively. The stock return is scaled by the sample standard deviation. In panel A, the variable *Wall Street* is a dummy variable that takes the value of one if the pickup or drop-off location is within 100 meters of Wall Street and zero otherwise. In panel B, we identify a business trip if the pickup or drop-off location is within 100 meters of listed companies between 7:30 a.m. and 6 p.m., and a personal trip if the trip is from 6 to 10 p.m. The variable *Personal trip* is a dummy variable that takes the value of one for personal trips and zero otherwise. Sample period is 2009–2016. The *p*-values adjusted by robust standard errors clustered by day and location are in parentheses.

*$p < 0.10$; **$p < 0.05$; ***$p < 0.01$.

listed companies from 7:30 a.m. to 6 p.m.) as personal trips. The reason is that there are many other unlisted firms in NYC whose headquarters we cannot track. Alternatively, we impose the further requirement for the taxi ride to occur during nonworking hours, for example, from 6 to 10 p.m.

We expect the tipping behavior of business trips to be less sensitive to market returns than that of personal trips, and we examine changes in the tipping behavior of the treatment group induced by stock market fluctuations in the following DiD specification.

$$\begin{aligned}\ln(Tip\ \ ratio)_{t,j} = \alpha &+ \beta_1 \times Personal\ \ trips_{t,i} \\ &+ \beta_2 \times Personal\ \ trips_{t,i} \times Return_t \\ &+ \gamma \times Controls_t + \varepsilon_t, \end{aligned} \tag{4}$$

where the dependent variable $\ln(Tip\ ratio)_{t,i}$ is the natural logarithm of the daily average tipping for group i on day t. The variable $Personal\ trips_{t,i}$ is our treatment indicator, which takes the value of one if the pickup or drop-off location is for personal purposes and zero if for business purposes. In the regression, we control for day fixed effects. The standard errors are clustered by day and trip purpose. The coefficient of interest is β_2 for the interaction term $Personal\ trips_{t,i} \times Return_t$, which captures the difference in real-time well-being induced by stock market fluctuations between the treatment and control groups. Panel B of Table 8 presents the results. We can see that across all empirical specifications, stock market fluctuations have a bigger effect for trips that are likely to be personal because the coefficients on the interaction of personal trips and the market return are significantly positive. This finding demonstrates that passengers who are more likely to be affected by market performance indeed exhibit a greater response in tipping behavior, providing support for the causal interpretation of our main results.

6. Additional Analysis

6.1. Stock Characteristics

Table 9 explores how stock characteristics can exaggerate or mitigate the influence of change in wealth on real-time utility. Our first test seeks to investigate portfolio and nonportfolio effects, where the relevant expectation that pertains to stock market fluctuations can contain information that can exert influence beyond an investor's portfolio, such as local income or job growth. Thus, we exploit geographic differences in firm headquarters by classifying firms as NYC based and non-NYC based and computing the daily value-weighted returns of the two groups of firms, respectively. The price movement for local firms would have both portfolio and nonportfolio effects, whereas non-NYC-based firms should primarily exhibit portfolio effects. The estimates for the two groups of firms are shown in columns (1) and (2) of Table 9. The point estimate for the most saturated specification is 27 bps (significant at the 10% level) for NYC-based firms. Continuous non-NYC returns alone can also influence the average tipping ratio, but the magnitude is similar, at 27.8 bps.

Does stock market capitalization play a role in affecting the documented association? Compared with smaller-cap stocks, large-cap stocks are more likely to gain attention because of media coverage. Pension fund investments tend to balance asset allocations toward stocks with larger market capitalizations. Thus, we separate stocks into two subgroups, large- versus small-cap stocks, based on the median daily market capitalization.[16] We then calculate the daily

Table 9. Impact of Stock Characteristics: Heterogeneity Test

	Portfolio return			
Dependent variable: ln(*Tip ratio*)	(1) *Non-NYC stock return*	(2) *NYC stock return*	(3) *Large stock return*	(4) *Small stock return*
Portfolio return	0.278**	0.270*	0.273**	0.026
	(0.04)	(0.05)	(0.05)	(0.84)
ln(*Total taxi fares*)	−0.772***	−0.772***	−0.772***	−0.774***
	(0.00)	(0.00)	(0.00)	(0.00)
Day-of-the-week fixed effects	Yes	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes	Yes
N	2,015	2,015	2,015	2,015
Adjusted R^2	0.127	0.127	0.127	0.125

Notes. The main independent variable in this table is the value-weighted daily portfolio stock return. The portfolio return *Non-NYC stock returns* is the value-weighted daily return of stocks with firm headquarters outside NYC. The portfolio return *NYC stock returns* is the value-weighted daily return of stocks with firm headquarters in NYC. The portfolio return *Large stock returns* is the value-weighted daily return of stocks with firms whose market capitalization is above median. The portfolio return *Small stock returns* is the value-weighted daily return of stocks with firms whose market capitalization is below median. All these stock returns are scaled by the corresponding sample standard deviation. The sample period is 2009–2016. The *p*-values adjusted by robust standard errors (White 1980) are in parentheses.

$^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

value-weighted returns for large- and small-cap stocks and reestimate Equation (1). The results are presented in columns (3) and (4) of Table 9. Consistent with our expectations, we find that the coefficient estimate for large-cap stocks is positive and statistically significant at the 5% level, whereas the impact on tipping behavior becomes small and statistically insignificant when we focus on small-cap stocks.

6.2. Default Option Bias

The behavioral finance and economics literatures have documented a large impact of default and heuristics on human behavior. Given the discretionary nature of tipping, we directly tackle the default-option bias in this section.

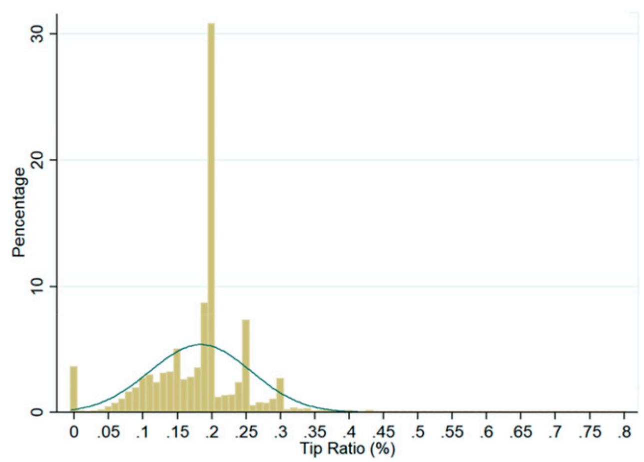
First, we test whether the market return affects passengers' behavioral biases in terms of clicking default app buttons for tipping. We present the histogram of the taxi tipping ratio at the trip level in Figure 3 and find evidence consistent with media reports about the psychological anchoring effects with respect to default options: sharp spikes are observed for the three default buttons for tipping, at 20%, 25%, and 30%, respectively.

Second, we examine the extent to which stock market fluctuations affect the deviation of taxi passengers' tips from the suggested default. We apply a similar specification as in Equation (1) to estimate the impact of the stock market return on day *t* on a default-option bias, where the default option is measured as the proportion of trips where passengers click the default tipping ratio, that is, 20%, 25%, or 30%. The results are reported in Table 10. We observe a very weak association between market returns and default options, no matter whether we examine each default option separately or at the aggregate level. Our results show that the proportion of passengers who click the default-option buttons are quite stable, and such a behavioral default is not affected by the stock market fluctuations that we document in the study.

6.3. Macro-Level Estimation

We now extrapolate these estimates to provide the aggregate magnitudes for gains and losses in tip revenue associated with stock market fluctuations for the taxi industry. The estimation is conducted according to the following procedure. First, we use 15%, which is viewed as common practice for tipping taxi drivers in NYC, as the benchmark ratio and multiply it by

Figure 3. (Color online) Taxi Tipping Distribution Histogram



Note. The *x*-axis is the taxi tipping ratio as a percentage, and the *y*-axis is the proportion of a given tipping ratio.

Table 10. Other Behavioral Biases: Default Options

Dependent variable: Proportion of default tipping	(1) Default option 20%	(2) Default option 25%	(3) Default option 30%	(4) Total proportion
Market return	–0.004	–0.002	0.005	–0.001
	(0.91)	(0.87)	(0.29)	(0.99)
ln(*Total taxi fares*)	0.007	0.002	–0.003	0.006
	(0.66)	(0.63)	(0.11)	(0.78)
Day-of-the-week fixed effects	Yes	Yes	Yes	Yes
Month × year fixed effects	Yes	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes	Yes
N	2,015	2,015	2,015	2,015
Adjusted R^2	0.985	0.970	0.964	0.984

Notes. The main independent variable in this table is the market return. The stock return is scaled by the sample standard deviation. The sample period is 2009–2016. The p-values adjusted by robust standard errors (White 1980) are in parentheses.

the aggregate fares to compute the regular tip revenue. In addition, to compute the market-induced tip revenue, we use both raw returns and those adjusted by the annual average to infer the tipping ratio. Finally, the gain/loss will be the difference between the tipping revenue in the first and second steps. Panels A and B of Table 11 report the implied gains/losses in tip revenue based on our high and low estimates in Table 2, respectively. Admittedly, these are plausibly conservative estimates of the true degree of aggregate tip gain/loss because they are computed based on the assumption that tipping occurs only once per year and occurs exclusively through card payment. As an alternative (more intuitive) approach to obtain the aggregate effects, we also produce a coarse estimate based on the results in Table 5, in which we differentiate the impact of positive and negative market returns. The results are tabulated in Table A4 of the online appendix. Across all years, we estimate that market-induced tip revenues range from –$17.515 million to $12.948

Table 11. Macro Estimation: Gains/Losses of Tips Associated with Stock Market Movement

Panel A: Using 0% as the benchmark

				Upper-bound coif. (0.293%)		Lower bound coif. (0.264%)	
Year	Aggregate fares (billions $)	Benchmark tip (millions $)	Annual market return	Induced tips (millions $)	Gain/loss of tips (millions $)	Induced tips (millions $)	Gain/loss of tips (millions $)
2009	0.473	70.950	31.02%	77.072	6.122	76.472	5.522
2010	0.643	96.450	17.99%	101.277	4.827	100.804	4.354
2011	0.810	121.500	–0.33%	121.388	–0.112	121.399	–0.101
2012	0.943	141.450	15.46%	147.533	6.083	146.937	5.487
2013	1.14	171.000	27.22%	183.948	12.948	182.679	11.679
2014	1.24	186.000	10.66%	191.516	5.516	190.975	4.975
2015	1.26	189.000	–0.56%	188.706	–0.294	188.734	–0.266
2016	1.19	178.500	12.89%	184.901	6.401	184.273	5.773

Panel B: Using the average annual market return, 14%, as the benchmark

				Upper bound coif. (0.293%)		Lower bound coefficient (0.264%)	
Year	Aggregate fares (billions)	Benchmark tip (millions)	Annual market return	Induced tips (millions)	Gain/loss of tips (millions)	Induced tips (millions)	Gain/loss – tips (millions)
2009	0.473	70.950	31.02%	74.250	3.300	73.926	2.976
2010	0.643	96.450	17.99%	97.440	0.990	97.343	0.893
2011	0.810	121.500	–0.33%	116.555	–4.945	117.040	–4.460
2012	0.943	141.450	15.46%	141.906	0.456	141.862	0.412
2013	1.14	171.000	27.22%	177.146	6.146	176.543	5.543
2014	1.24	186.000	10.66%	184.117	–1.883	184.301	–1.699
2015	1.26	189.000	–0.56%	181.187	–7.813	181.953	–7.047
2016	1.19	178.500	12.89%	177.800	–0.700	177.868	–0.632

Note. The upper/lower bound of elasticity of tips to wealth change is obtained from the coefficient estimates in columns (2) and (3), respectively, of Table 2. Coif, coefficient.

million. If we look at specific years, we can calibrate this magnitude on a per capita basis. For example, in 2010, total fares were $643 million, which, according to our beta estimate in Table 2, corresponds to a tip revenue gain between $0.990 million and $4.827 million. This amount, for the total of 49,129 licensed drivers reported by the TLC in 2010, implies an extra gain of $20.15 to $98.25.[17] Given NYC taxi drivers' mean annual salary in the same year, $30,650, the stock market performance implies a salary increase of 0.06%–0.32%.[18]

7. Conclusions

Prospect theory assumes that utility depends on wealth losses/gains rather than on the levels and individual value losses and gains in the decision-making process. Exploring a unique data set of NYC taxicab tipping records, we develop a novel and symmetric utility measure to capture variations in both excitement and discontent and study how changes in investor wealth affect real-time utility.

Our key results are summarized as follows. We find that the sensitivity of the tipping ratio to stock market performance is 0.3, indicating that a one-standard-deviation (~1.1%) decline in the stock market increases the daily average tipping ratio by 0.3%. The impact is nearly instantaneous and is driven by losses rather than gains in wealth. The effect is robust to controlling for a series of checks and consistent with a wealth-effect interpretation. We further conduct two DiD tests by exploring the extensive coverage of NYC and high-frequency nature of the data. The results support causal inference. Finally, we find that the effect is particularly strong for the stocks of firms with large market capitalizations.

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Appendix. Variable Definitions

Variable	Definition
ln(*Tip ratio*)	The natural logarithm of daily average tip ratio. The daily average tip ratio is defined as (the sum of tips divided by sum of total fares) × 100 based on the trip data after trading hours starts (9:30 a.m.) in a given day.
ln(*Total taxi fares*)	The natural logarithm of the average daily taxi fare. The average daily taxi fare is defined as the average total fares for each pickup.
Market return	The value-weighted daily return of U.S. stocks
S&P 500 return	The value-weighted daily return of Standard & Poor 500 stocks
Negative return	The stock return if the stock return is negative and zero otherwise
Positive return	The stock return if the stock return is positive and zero otherwise
NYC stock return	The value-weighted daily return of stocks with firm headquarters in NYC
Non-NYC stock return	The value-weighted daily return of stocks with firm headquarters outside NYC
Large stock return	The value-weighted daily return of stocks with firms whose market capitalization is above median
Small stock return	The value-weighted daily return of stocks with firms whose market capitalization is below median

Endnotes

[1] Changes or differences are referred to as gains and losses over which the utility function or value function is defined.

[2] Recent theoretical studies include those of Koszegi and Rabin (2006, 2007, 2009) and Hsee and Tsai (2008).

[3] For example, in the famous World Values Survey, each respondent is asked the following question: "Taking all things together, would you say you are ____." Responses are categorized using a quantitative happiness score. Krueger et al. (2009) collected information on individuals' self-evaluations of their emotional experiences in their various uses of their time in a process called *evaluated time use*.

[4] A sizable literature estimates how consumer spending responded to the federal income tax rebates or an unanticipated income shock (e.g., Souleles 1999, Johnson et al. 2006, Agarwal and Qian 2014).

[5] Tipping is also an important economic phenomenon. Estimated tips in U.S. restaurants alone total about $27 billion annually (Azar 2004). This figure is higher when tips in other establishments, such as hotels and taxis, are considered, especially in other countries.

[6] See https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page.

[7] Tipping has received attention from researchers in such diverse disciplines as economics, psychology, services marketing, and tourism (e.g., Lynn and Grassman 1990, Ruffle 1999, Azar 2010, Gneezy et al. 2010).

[8] Each vehicle affiliated with the Boro Taxi Program has the same metered fare rate structure as the yellow taxis. The new program allows yellow taxicabs to pick up passengers anywhere in the five boroughs but imposes constraints on pickup areas for the Boro taxis. These are only allowed to pick up passengers off the street in northern Manhattan (north of West 110th Street and East 96th Street), the Bronx, Queens (excluding LaGuardia and John F. Kennedy International (JFK) airports), Brooklyn, and Staten Island but not in the exclusive Manhattan zone. However, Boro taxis can drop off passengers anywhere. Enforcement of the program requires (1) that all Boro taxi drivers have their cars painted and the taxi logo and

information printed, with the affiliated base indicated on the rear and sides of the car, and cameras, meters, and GPS installed, and (2) that the in-car GPS not allow the meter to work if the cab is starting in Manhattan below East 96th Street or West 110th Streets or at the LaGuardia and JFK airports (TLC 2014).

[9] The TLC data provides the taxi zone ID instead of GPS information since July 1, 2016. Thus, in some tests using GPS information to identify the pickup/drop-off location, we use the sample period until June 30, 2016.

[10] Delayed reactions are not an issue in our setting because the impact on real-time well-being or happiness is expected to be contemporaneous.

[11] We provide additional robustness tests on the relation between market fluctuations and taxi tipping in Table A1 of the online appendix. Columns (1) to (3) of Table A1 report the results when the default tip ratio (tips that are exactly or very close to 10%, 15%, or 20%) are excluded, whereas column (4) and (5) show the estimate when number of rides and route fixed effects are added to the specification, respectively. The findings remain robust.

[12] Prospect theory cannot use a logarithm as a dependent variable. Running the regression with logarithms changes the shape of the curve being fitted, making it difficult to interpret within the context of prospect theory. To further address concerns about the magnitude of the effect resulting from the logarithm of the tipping ratio, we also repeat the regression specification using the tipping ratio itself as the dependent variable. Table A2 in the online appendix shows that the overall findings remain significant.

[13] The TLC only releases the vendor(s), Creative Mobile Technologies and Verifone, that provide the record.

[14] As shown in Table A3 of the online appendix, there are no differences in terms of number of passengers between Wall Street and non–Wall Street trips, but trips from/to Wall Street area are more likely to be paid for by credit card, have longer trip distances, and are less likely to receive zero tips.

[15] In an unreported test, we show that the result is robust to the exclusion of ln(*Total taxi fares*) as a control variable.

[16] As a robustness check, we also refine the stock size split according to Fama–French New York Stock Exchange breakpoints.

[17] See https://www1.nyc.gov/assets/tlc/downloads/pdf/annual_report_2010.pdf.

[18] See https://bizfluent.com/info-8496448-much-make-new-york-city.html.

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